# Helia Vafaei

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## LINKS

LinkedIn, GitHub, Personal website

## EDUCATION

Sep 2025 – Present

**M.Sc. in Information Technology (IT) Engineering**
**Major: Networked Systems**
**University of Tehran** Tehran, Iran
Among the top 2 universities in Iran according to QS University Rankings

Sep 2020 – Feb 2025

**B.Sc. in Computer Engineering**
**Iran University of Science and Technology** Tehran, Iran
Among the top 4 universities in Iran according to QS University Rankings
GPA:

- Last two years: **3.58/4.00** (**17.56/20.00** on Iranian Scale)
- Overall: **3.58/4.00** (**17.27/20.00** on Iranian Scale)

## RESERCH INTERESTS

**Software Engineering**
**Software Development**
**Data Engineering**
**Data Science**
**Data Analytics**
**Software Verification and Testing**
**Programming Languages and Compilers**

## SKILLS

**Programming Languages:**
Python, React, HTML, CSS, JavaScript, C++, C#, C, SQL

**Libraries and Frameworks:**
React, Bootstrap, .Net, Keras, TensorFlow, Numpy

**Environmental and Tools:**
Git, Azure Devops, Linux, Bash, Postman

## SELECTED PROJECTS

**NoWaste:**

- Participated as a Front-end developer in developing NoWaste, a web application aimed at minimizing food waste by connecting restaurants and cafes with surplus food to potential consumers.

- Designed and implemented RESTful APIs to facilitate efficient data exchange between frontend and backend systems.
- Dockerized the application to streamline deployment and ensure consistency across development and production environments.
- Developed and maintained CI/CD pipelines, automating testing, building, and deployment processes to enhance development efficiency and ensure robust application delivery.

**Anti-spoofing algorithm:**

- The project was for checking whether a face in the video is alive or not.
- In this project, two different models, including a feature-based and a deep learning model, were used.
- For the feature-based model, SVC was used.
- For the deep model, VGG16 and some extra un-pretrained layers were used.
- The data came from CASIA-FASD.

**Computer system security project:**

- The main aim of this project is retrieving information from the victim device using Bash in Linux and displaying it in a simple website.

**Software quality measurement and its quantification:**

- By running this project, the quality of a software can be measured quantitatively. In the software quality article, factors are presented that depend on a number of criteria, but the criteria is qualitative, so we use a number of metrics to achieve a quantity.

## PUBLICATIONS

**Intelligent Resource Allocation in IoT: An ACO-GA Approach**

- Publisher: IEEE
- Advisor: Dr. Marzieh Malekimajd
- Co-author: Negin Haghighi
- Summary: This research introduces a hybrid approach that integrates Ant Colony Optimization (ACO) and Genetic Algorithm (GA) to enhance resource allocation in Internet of Things (IoT) environments.

## HONORS AND CERTIFICATES

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|---|---|
| Oct 2025 | Credit Risk Modeling in Python, DataCamp |
| Oct 2025 | ETL and ELT in Python, DataCamp |
| Oct 2025 | Exploratory Data Analysis in Python, DataCamp |

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| Oct 2024 | Ranked within **the Top 10% among about 100** in my class in GPA at IUST |
| Oct 2023 | Programming for Everybody (Getting Started with Python), Coursera |
| Jul 2022 | Algorithms on Strings, Coursera |
| Mar 2022 | Algorithms on Graphs, Coursera |
| Dec 2021 | Data Structures, Coursera |
| Nov 2021 | Algorithmic Toolbox, Coursera |

## ACADEMIC EXPERIENCE

**Honorary Member**

- Voluntary work in Computer Engineering Scientific Association at Iran University of Science and Technology (CESA IUST) – Fall 2023

**Teaching Assistant**

- Systems Analysis and Design – Instructor: Dr. Mehdad Ashtiani – Fall 2024
- Data Transmission – Instructor: Dr. Marzieh Sheikhi – Spring 2024
- Artificial Intelligence – Instructor: Dr. Arash Abdi – Fall 2023
- Database Design – Instructor: Dr. Hossein Rahmani – Spring 2023
- Fundamentals of Computer Programming – Instructor: Dr. Marzieh Davoodabadi Farahani – Fall 2022
- Advanced Computer Programming – Instructor: Dr. Sauleh Eetemadi – Spring 2022
- Advanced Computer Programming – Instructor: Dr. Sauleh Eetemadi – Fall 2021

## LANGUAGES

**Persian:** Native Proficiency
**English :** Full Professional Proficiency IELTS (Academic): 7.0 (Overall score)